Supplementary Information

Geometric Constraints on Transition Dynamics in Recursive Language Systems

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Supplementary summary

This document contains extended methodological notes, robustness figures, initial-condition analyses, a short references section, and a brief AI-use disclosure supporting the main manuscript.

S1. Overview

This supplement provides methodological detail, robustness analyses, and extended figures supporting the main manuscript. The central aim is to clarify how Fisher geometry, phase-boundary proximity, curvature, and phase-sensitive observables jointly constrain transition behavior.

S2. Fisher-geometric construction

State space. We construct a state space over recursive outputs using parameterized probes indexed by control parameters *(r, α)* and associated observables.

Metric. A Fisher information metric defines a Riemannian manifold over this parameter space [1–4]. Distances between states are derived from this metric and used to construct a weighted graph.

Embedding. Multidimensional scaling (MDS) is used only for visualization [5]. Primary observables—distance to phase boundary, curvature proxies, transition labels, and Lazarus-related summaries—are computed before dimensionality reduction.

Phase boundary. The phase boundary is estimated from curvature and transition structure and serves as the primary geometric separator between regimes.

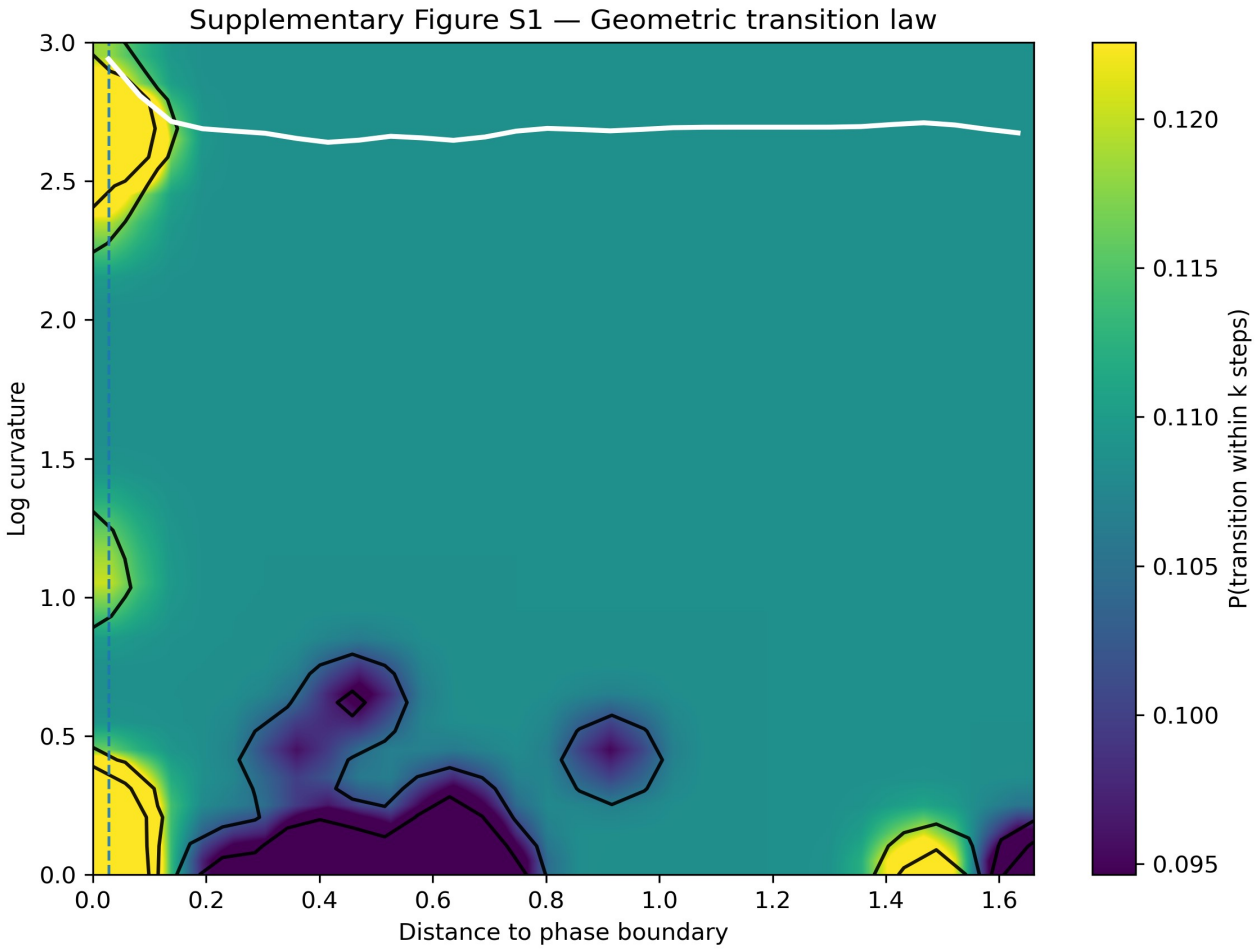
S3. Geometric mechanism

The empirical results support a hierarchical picture. Distance to the phase boundary determines whether transitions are accessible at all, while curvature modulates transition likelihood within that accessible region. In this sense, boundary proximity acts as a gate and curvature shapes local expression.

A natural summary form is a conditional law of the form *P(transition | geometry)*, where geometry is represented by boundary distance and curvature. The data do not support collapse onto a single global scalar; instead, the system exhibits piecewise geometric regimes.

S4. Supplementary Figure S1: geometric transition law

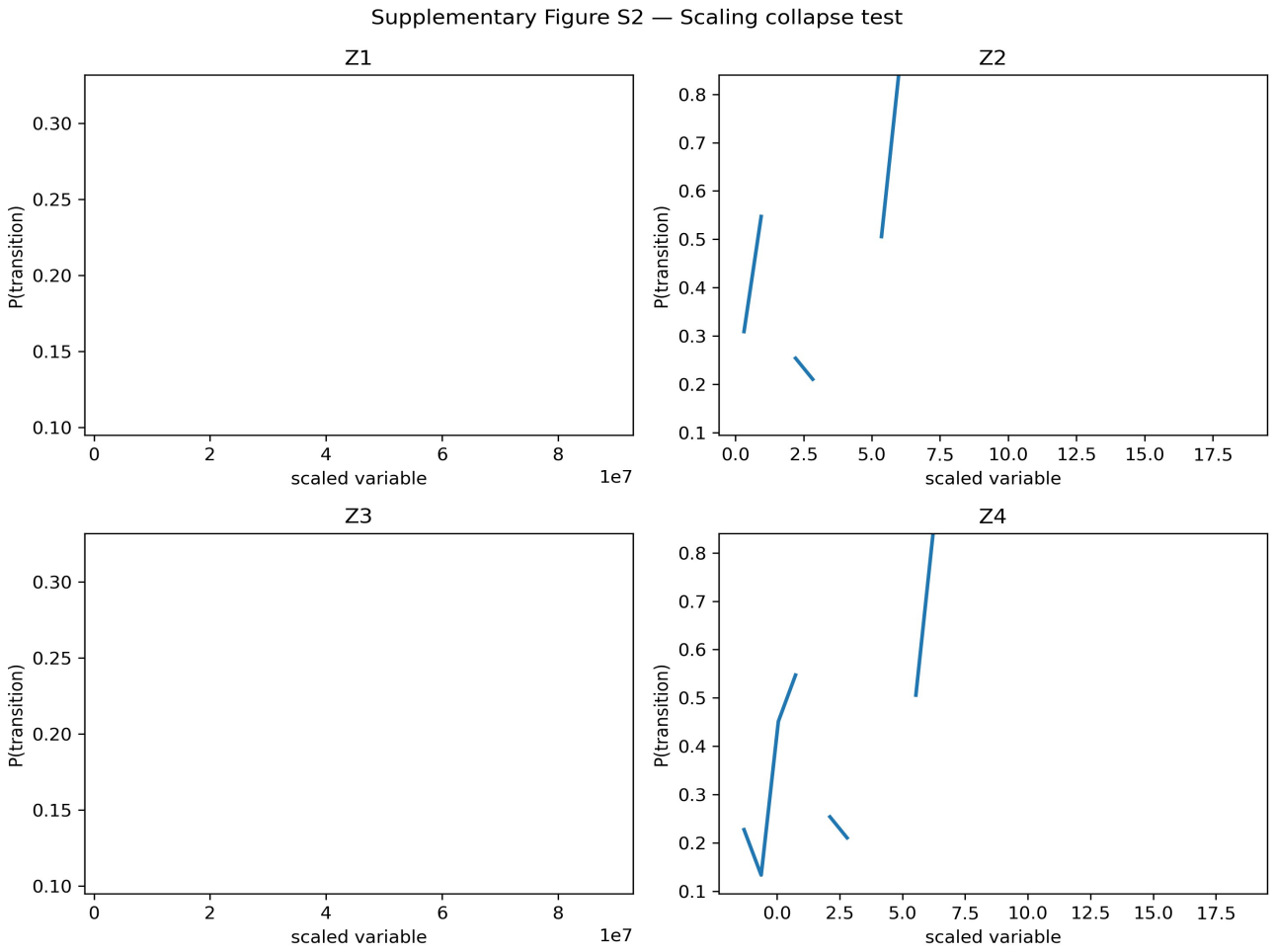
Supplementary Figure S1 visualizes transition probability as a geometric field over distance to phase boundary and log curvature. Transitions concentrate along a boundary-adjacent ridge, with curvature amplifying transition likelihood within that region.



Supplementary Figure S1 — Geometric transition law. Transition probability as a function of distance to phase boundary and log curvature. Probability concentrates along a boundary-adjacent ridge, while a marginal distance profile confirms that boundary proximity remains the primary organizing coordinate.

S5. Supplementary Figure S2: failed global collapse

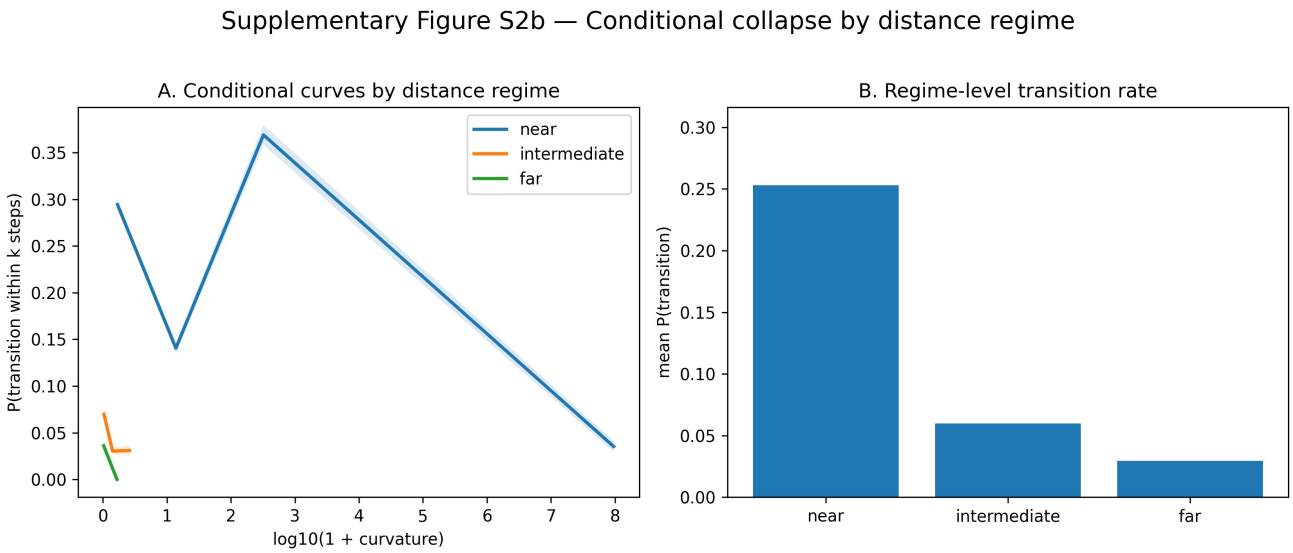
We tested whether transition probability collapses onto a single effective coordinate combining curvature and boundary distance. No clean global collapse was observed, indicating that transition dynamics are not reducible to a naive one-dimensional scaling variable.



Supplementary Figure S2 — Global scaling collapse test. Several candidate one-dimensional scalings of curvature and boundary distance fail to produce a clean global collapse. This indicates that transition dynamics are not reducible to a single naive scalar coordinate.

S6. Supplementary Figure S2b: conditional collapse by regime

Conditioning on distance to the phase boundary reveals the missing structure. Near-boundary states exhibit a strong curvature-dependent transition law, intermediate states show a weaker response, and interior states remain dynamically suppressed. The absence of a global collapse is therefore informative: the system is organized into regime-dependent geometric laws.

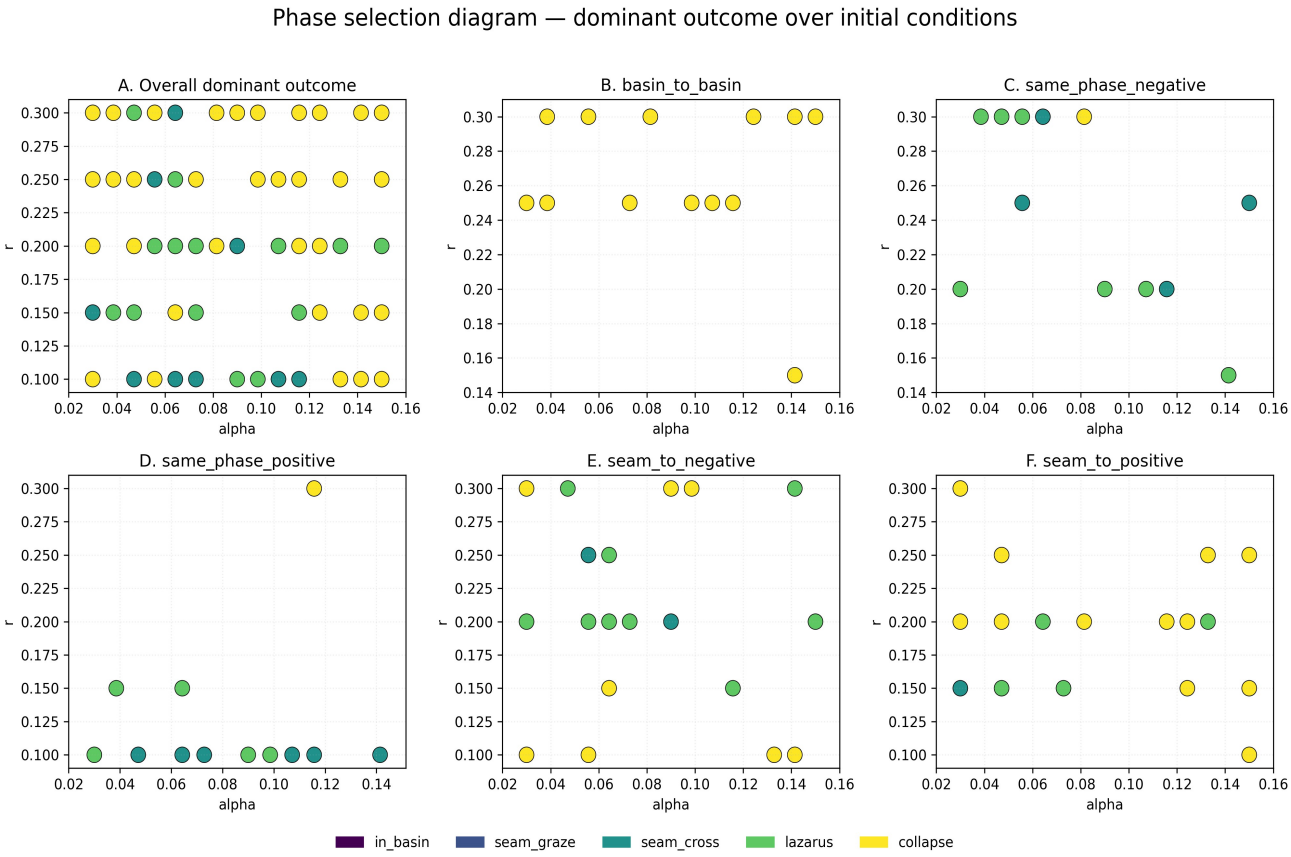


Supplementary Figure S2b — Conditional collapse by distance regime. When stratified by boundary distance, transition curves reveal distinct geometric regimes: strong curvature-sensitive dynamics near the boundary, weaker intermediate response, and suppression in the interior.

S7. Initial conditions and phase selection

We next classify trajectories by initial-condition family and map their dominant outcomes over parameter space. Outcomes are not sampled randomly. Instead, the manifold partitions into selection regimes governing whether trajectories collapse, cross the phase boundary, or enter Lazarus-like behavior.

This selection view extends the main manuscript from geometric description to predictive behavior. Geometry constrains what can happen; initial conditions determine which outcomes are realized.



Supplementary Figure S3 — Phase selection diagram. Dominant outcomes over initial-condition families and parameter space. Behavioral classes are clustered rather than random, indicating that the manifold partitions into structured selection regimes.

S8. Reproducibility and pipeline sealing

Canonical refresh scripts ensure that figure-facing CSVs are regenerated from current trajectory outputs before manuscript production. Default scale coverage and pre-push sealing are used to reduce drift between raw outputs, summaries, and figures.

S9. AI use disclosure

A large language model was used to assist with language development, structural editing, figure-caption drafting, and manuscript refinement. All scientific decisions, analyses, interpretations, and final wording were reviewed and approved by the author, who takes full responsibility for the content.

S10. Outlook

The present results motivate a more explicit theory of transition ridges and selection regimes on information-geometric manifolds. A next step is to derive these structures directly from the Fisher metric and to test whether the same boundary-conditioned laws recur across other recursive or generative systems.

Supplementary references

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